

Review

The potential of machine learning in diagnosing neurological and psychiatric diseases: a review

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Abstract

Purpose of the study Artificial intelligence (AI) is rapidly transforming medical practice, from patient care to diagnosis and treatment personalization. With our work, we aim to explore the application of machine learning (ML) and deep learning (DL) algorithms in the diagnosis of neurological and psychiatric diseases, exploring into both the benefits and the challenges associated with these new technologies.

Findings The examined technologies have shown considerable success in early diagnosis, as well as in the identification of risk factors and symptom management for diseases such as Alzheimer's, Parkinson's and psychiatric disorders. These models could help improve diagnostic accuracy and enable a more personalized therapeutic approach by utilizing large datasets with information such as biomarkers and medical images. However, certain challenges persist, including concerns about data quality, patient privacy and the ethical implications of algorithmic decisions.

Summary Artificial intelligence-based diagnostic methods offer great potential to enhance early diagnosis and, consequently, the management of neurological and psychiatric disorders. To maximise their application, it is essential to ensure transparency and interpretability of the models, which are fundamental for their safe and effective use in medical practice.

Keywords Artificial intelligence · Machine learning · Deep learning · Diagnosis · Neurological diseases · Psychiatric disease

1 Introduction

The recent advent and rapid development of artificial intelligence (AI) are transforming medical practice. Similar to many other research fields, healthcare has witnessed unprecedented developments over the last decade. The ability to personalise patient care, assist clinicians in decision-making, and enhance treatment effectiveness are all areas in which artificial intelligence has demonstrated significant value [1, 2]. Specifically, machine learning enables the extraction of latent patterns directly from data, which are then used to make inferences. This approach is further enhanced by the vast amount of data produced by the healthcare systems, insurance companies and medical research institutions [3]. Access to this enormous amount of Big Data allows Artificial Intelligence tools to improve their performances in order to optimize the decision-making process, formulate early diagnoses and personalize treatment by predicting the possible outcome [4]. Beyond effectiveness, these methods also raise numerous ethical issues. Debates related to patient privacy, data security, accountability for decisions and respect for individual

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autonomy are emerging and are fundamental to the correct implementation of these technologies [5]. Thus, several new challenges must be addressed to facilitate the full integration of these technologies into the healthcare systems. For instance, the development of interpretable systems is fundamental: understanding how algorithms extract data and create outputs is crucial for the usability and acceptance of such tools [6].

Among all the possible tasks that AI can handle in medical practice, particular importance is given to its role in diagnosis. Machine learning and deep learning technologies have been employed for years to differentiate healthy subjects from patients or to predict the disease progression. This approach has found success in various branches of medicine, including oncology [7], neurology [8], radiology [9] and psychiatry [10]. In each case, algorithms handle data such as biomarkers [11] or medical scans [12], identifying hidden patterns to infer the patient's health status. Therefore, these tools are essential for obtaining a rapid and early diagnoses, ensuring unbiased decision-making and improving symptoms management. Regarding diagnostic accuracy, in recent years, several studies have compared AI to clinicians [13] [14]. For instance, Kim et al. (2020) [15] shows how their AI model is more sensitive than clinicians in diagnosing breast cancer with mammography (0.9 vs 0.78), even in the case of early breast cancer (0.91 vs 0.74). Again, Becker et al. (2022) [16] evaluated the ability of a deep learning model to identify pneumonia in chest radiographs and then compared it to the ability of two radiologists. As a result, AI outperformed the clinicians: the AI model's sensitivity and specificity were 95.4% and 66.0% respectively, while the clinicians achieved only 50.6% and 73%. Also, Esteva et al. (2017) [17] evaluates the performance of a CNN trained to classify skin cancer and compares it to two dermatologists. The AI model reaches an overall accuracy of $72.1 \pm 0.9\%$ with a three-class disease partition (versus 65.56% and 66.0% accuracy of the clinicians) and $55.4 \pm 1.7\%$ with a nine-class disease partition (versus 53.3% and 55.0%).

In recent years, huge strides in the use of artificial intelligence for diagnosis have been made in neurology and psychiatry, where many models have been developed with different goals. For instance, Badža et al. (2020) [12] proposes a new Convolutional Neural Network architecture to classify three brain tumor types: meningioma, glioma, and pituitary tumor. The tested model, trained using a database composed by T1-weighted contrast-enhanced MRI images, showed good generalisation capability and reduced execution speed, while maintaining a simple architecture. Regarding neurodegenerative disease detection, Gupta et al. (2023). [18] shows how AI can promote Parkinson's disease early diagnosis. Machine learning models have proven useful to identify diseased patients using different types of data. Speech recordings can be used to automatically discover vocal issues, while handwriting characteristics are used to identify bradykinetic movements and movement control. At the same time, gait data have been used to differentiate between healthy controls and Parkinson's individuals. Finally, another source of data useful for this task are imaging scans, where machine learning models can automatically find abnormalities in the scans to diagnose Parkinson's disease. AI has proven useful also in other neurological pathologies diagnosis, such as epilepsy. For instance, Yuan et al. (2018) [19] proposes a new method to automatically identify epileptic EEG signals. This procedure relies on both a Support Vector Machine classifier and Local Binary Pattern Operators and reaches high levels of recognition accuracy. Finally, AI has been used to identify psychiatric disorder subtypes. For instance, Zhang et al. (2021) [20] shows the identification of two clinically relevant subtypes of both post-traumatic stress disorder and major depressive disorder, discovered thanks to a machine learning procedure based on high-density resting-state electroencephalography.

In summary, this paper aims to investigate the current state of the literature on the use of machine learning and deep learning algorithms in the diagnosis of both neurological and psychiatric diseases. We will highlight from the existing literature the benefits and the challenges associated with these methodologies, examining into the most relevant approaches and analysing the most used algorithms and metrics, starting from the most cited case studies.

The paper is structured as follows (Fig. 1): Sect. 1 offers a general overview of the use of machine learning algorithms to diagnose both neurological and psychiatric diseases. Section 2 contains a brief and general introduction to the concepts of machine learning and deep learning. Section 3 reviews the most significant existing literature on this topic. Not all the papers analysed in this section have purposes similar to ours, but they still offer an in-depth analysis of the pros and cons of using artificial intelligence in diagnosis. In Sect. 4 we outline the methodological criteria that guided the construction of the dataset and the formulation of the research questions. Section 5 presents the results of our analysis, addressing the first four research questions. Section 6 corresponds to the analysis of the ten most cited papers. In this section we will explore: the most used algorithms, which metrics have been used to evaluate their performance, and

which challenges have been addressed. In Sect. 7 we discuss the findings and present commentary on the results, while in Sect. 8 we outline potential limitations and future directions. Section 9 serves as a brief conclusion.

The need to develop this work arises from the growing incidence of neurological and psychiatric disorders worldwide, which in turn increases the urgent demand for accurate and timely diagnostic solutions. Current clinical approaches

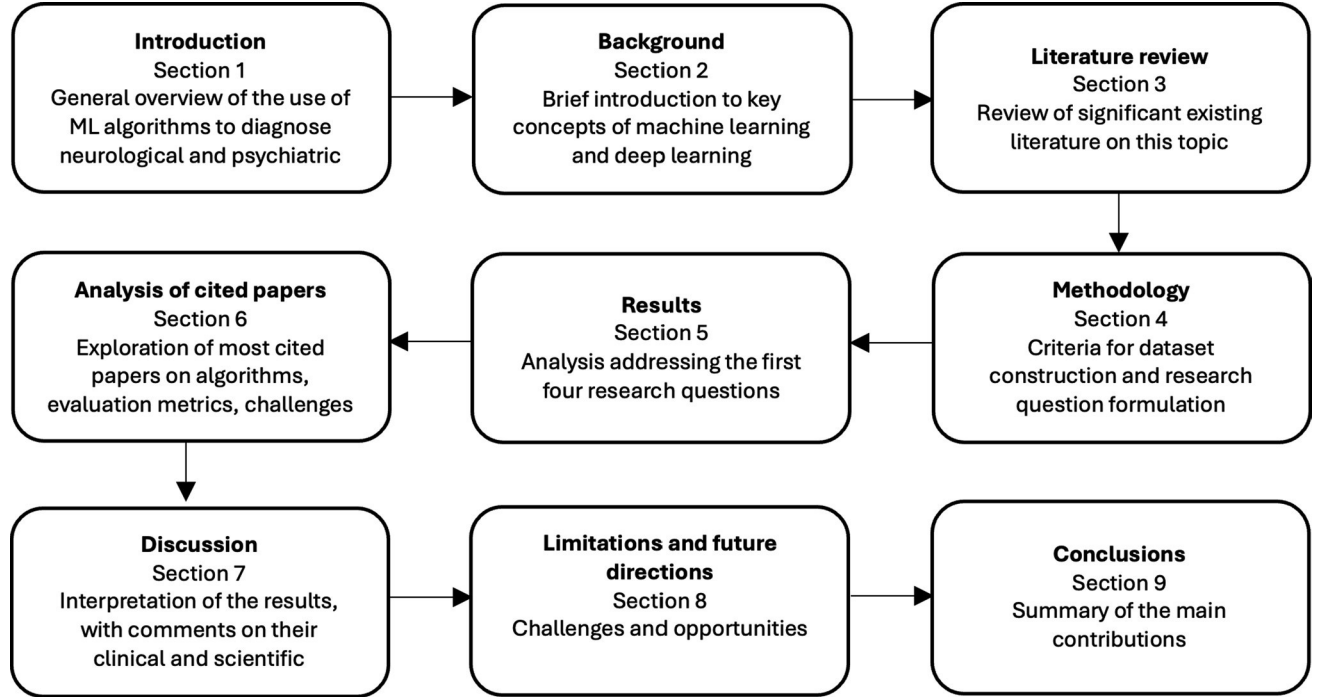


Fig. 1 Graphical Representation of the Article Index

are often imprecise and time-consuming. For this reason, we chose to highlight the potential of machine learning in enhancing diagnostic capabilities through the analysis of large-scale data. The main contributions of this work can be summarized in three key points: first, it provides an overview of recent applications of machine learning in the field of neurological and psychiatric disorders; second, it analyzes the performance and limitations of the various algorithms and data modalities used in these studies. Finally, it identifies the main challenges and future directions, serving as a guide for integrating ML tools into clinical practice. This study emphasises both the technical and clinical aspects of machine learning, offering a multidisciplinary perspective aimed to bridging the gap between data science and medicine.

2 Machine learning and deep learning

Before exploring into the use of AI-driven diagnostic methods in medicine, it is appropriate to explain the concepts underpinning our research. Machine learning is a discipline that arises from the intersection of statistics and computer science and deals with the study of how computers can learn directly from data, whether these data consist of numerical values or images. Through this process, algorithms can be trained to accomplish a variety of tasks. According to the type of training, ML can be categorised in supervised learning, unsupervised learning and reinforcement learning [21]. Supervised learning entails training an algorithm to produce a known output. This requires the use of “labelled” data, which include both the features and the correct outputs. In this way, the algorithm’s performances can be verified by comparing the predicted output to the desired output, thus determining whether further training is necessary. Supervised learning is particularly useful to make inferences about new, unseen data, thanks to the knowledge gained in the training phase [22]. Conversely, unsupervised learning involves training the system on unlabelled data. This enables the model to uncover latent patterns within raw data without the need for explicit instruction [23]. Lastly, reinforcement learning represents a hybrid approach, utilising a reward-and-punishment mechanism. Based on the accuracy of its outputs, the model is rewarded or punished in order to optimize future inferences. Thanks to the feedback it receives, the model improves its performances, providing more accurate outputs [24]. Machine learning algorithms can be employed

for both classification and prediction tasks and, depending on the task they were trained for, they have proven useful across various objectives. Over time, numerous machine learning algorithms have been developed, each with distinct mechanisms. Linear regression, logistic regression, decision trees, support vector machines (SVM) and K-nearest neighbors (KNN) are among the most widely used algorithms, which will be examined in greater detail in Sect. 6 [25].

Deep learning, a specialised subset of machine learning, utilises artificial neural networks (NN) to handle data, identify complex patterns and make predictions. A deep neural network (DNN) comprises multiple layers, each of which work on the representations it receives from the preceding layer and transforms them before passing the output to the next layer. Each layer consists of a set of nodes, which function as basic processing units inspired by biological neurons. Data processing starts at the input layer and propagates forward through subsequent layers. Adjustable parameters, referred to as “weights”, are applied to the input at each layer and are subjected to activation functions that introduce non-linearities. In this way the network can progressively process at a higher level of complexity and abstraction. To ensure accuracy and enhance network performance, the output created by the network is compared against the desired output, and the error is calculated using a loss function. A reduction in the loss function corresponds to improved output accuracy. To achieve this, the error is propagated backwards through the layers, where the weights are adjusted. This iterative procedure is repeated until the desired performance level is attained [26]. The ability to modify their architecture and parameters, coupled with their high effectiveness, has rendered neural networks exceptionally useful for executing complex tasks by processing vast quantities of raw data. Their adaptability and performance have solidified their role as indispensable tools in addressing challenging problems across numerous domains.

3 Related works

In the existing literature, several reviews already address the implementation of AI tools for the diagnosis of both neurological and psychiatric disorders. While these were not included in our analysis, this section offers a general overview of the most relevant reviews.

Dwyer et al. (2018) [10] explores the potential future developments facilitated by the implementation of AI in the diagnosis, prognosis and treatment of psychiatric disorder. This integration has the potential to optimise diagnostic processes, increase the generalizability of results and reduce decision-making time. Furthermore, ML autonomously learns methods and parameters, leading to unbiased and highly generalizable results. Consequently, the limitations inherent in traditional statistical methods are avoided. However, the authors highlight that this approach also has its own limitations that must be addressed in the future: from the algorithms’ explainability to the ethical implications related to their clinical implementation, from the validity of existing diagnostic and prognostic labels to the importance of using training data that are truly representative of the population.

Regarding the implementation of AI in psychiatry, Bzdok et al. (2017) [27] focuses on a different aspect: machine learning algorithms allow treatment predictions to be extrapolated directly from data. Rather than relying on pre-defined disease categories and hypotheses, these new methods are completely data-driven and can outperform conventional statistical techniques. In this way, a truly personalised and specialised medicine can be built, based on individual-level data, rather than group-level statistics that require formal hypothesis testing. More deeply, psychiatric disorders involve complex mechanisms ranging from molecular to social factors, making it challenging to formulate comprehensive hypotheses. ML, however, can detect disease-specific biological patterns, providing an opportunity to better address pathologies which are rooted both into genetics and life experiences.

Shatte et al. (2019) [28] present a broad overview of machine learning applications in mental health, focusing on a practical point of view. The results obtained demonstrate that AI can be applied across four domains: diagnosis, prognosis and treatment, public health, and research. In particular, the literature focuses predominantly on diagnosis, with more emphasis on depression, Alzheimer’s syndrome and schizophrenia. However, it is highlighted how other pathologies and other areas are less explored. Overall, ML techniques have proven extremely effective in improving clinical and research processes. Among these, classification techniques, particularly for diagnostic purposes, are the most explored, while new unsupervised techniques have not yet been meagerly investigated.

Low et al. (2019) [29] focuses on the possibility of introducing speech processing technologies to improve the diagnosis and treatment of mental illnesses. Currently, diagnostic methods rely on the evaluation of patient reports, which can be influenced by subjective biases, cognitive deficits and social stigma. However, emerging technologies, including ML

algorithms, allow to extrapolate behavioral and biological data in an unbiased manner, improving the objectivity of psychiatric assessment. This review synthesises findings from studies that employ automatically extracted acoustic features to detect psychiatric disorders and try to identify acoustic features that significantly differ in psychiatric disorder compared to neurotypical populations, linking them to the symptoms. Furthermore, it offers guidelines for data acquisition and modeling to enhance reproducibility and generalizability of results.

Focusing on the case of depression, Gao et al. (2018) [30] examines ML-based classification and prediction studies of major depression disorder (MDD) using features derived from MRI data. Altered brain networks can thus be utilised to discriminate patients from healthy individuals and to predict the outcomes of every possible treatment. Despite its promise, this approach faces challenges: the small sample sizes, the high feature dimensionality and the risk of overfitting the data.

There are many other studies addressing the application of ML not only in mental health, but also in diagnosing neurological disorders.

For instance, Raghavendra et al. (2019) [31] reviews the most widespread machine learning methods for the diagnosis of five neurological diseases: epilepsy, Parkinson's disease, Alzheimer's disease, multiple sclerosis, and ischemic brain stroke. The authors highlight the increasing prevalence of computer-aided diagnosis (CAD) systems in clinical practice which support clinicians in diagnostic and decision-making processes.

Myszczyńska et al. (2020) [32] highlights the importance of utilising ML in diagnosing and treating neurodegenerative disorders. Unlike human analysts, algorithms can detect patterns and make inferences from multidimensional data. This enables early diagnoses from neuroimaging data, motor degeneration metrics and genetic analyses. On the other hand, the authors highlight how these new methods can also be used to develop new and effective therapies.

Battineni et al. (2020) [33] discusses the slow progression of chronic diseases and the resulting need for early diagnosis, in which artificial intelligence can be a valuable tool. By identifying the correct decision model, it is possible to diagnose chronic diseases and predict patient outcomes. ML predictive models are widely utilised in disease diagnosis; however, there is no standard method to define the best approach in clinical practice. Among the models considered are support vector machines (SVM), logistic regression (LR), and clustering. These models are expected to become increasingly valuable in future clinical practice, aiding decision-making and autonomous diagnoses in accordance with established regulations. Specifically, in nervous system disorders, the linear regression model has achieved an accuracy between 72 and 80% in identifying the causes of depression.

Finally, Shoeibi et al. (2021) [34] reviews methods used for automatic epileptic seizure detection through DL techniques. Since 2016, this research field has seen a significant increase, utilizing DL models such as convolutional neural networks (CNN), recurrent neural networks (RNN), deep belief networks (DBN), autoencoders (AE), CNN-RN and CNNAE. Among these, CNNs are employed for disease diagnosis using biological signals, thanks to their mono or bidimensional architectures. In particular, 2D-CNN and 1D-CNN models are the most widely used for epileptic seizure detection. Recently, a novel 2D-CNN model was developed, capable of extracting both spectral and temporal features of EEG signals to learn the general structure of seizures [35]. However, it is not possible to identify a DL model superior to others for epileptic seizure detection. Thus, the structure must be chosen based on the dataset and the characteristics of the problem.

4 Research methodology

This section outlines the research methodology employed in this study, divided into five main phases, which can be summarized as follows: (4.1) formulation of research questions, (4.2) data source and search strategy, definition of (4.3) inclusion and (4.4) exclusion criteria, (4.5) data preprocessing, (4.6) data analysis and visualization, and (4.7) evaluation of performance metrics.

5 Research questions

First, the research questions that this paper aims to answer were defined:

RQ1 How many scientific studies have been published between 2018 and 2024 regarding the use of AI-driven diagnostic methods?

Objective: to determine the volume of research and analyse the evolution of studies on ML diagnostic algorithms.

RQ2 What are the most relevant research centers for studies on AI for the diagnosis of mental and neurological disorders?

Objective: to identify the primary sources of publication in this field.

RQ3 Which countries had the most active research centers?

Objective: to map the geographical distributions of research activity.

RQ4 In which disorder is AI most commonly applied

Objective: to identify the primary disorder where AI is frequently utilised.

RQ5 Which AI algorithms have been most commonly used to diagnose both neurological and mental disorders?

Objective: to analyse the predominant algorithms applied in this research field.

RQ6 What metrics were used to evaluate performance of the identified AI tools?

Objective: to determine the criteria used to measure the effectiveness and efficiency of the most commonly used AI tools.

RQ7 What challenges have been addressed in the development and implementation of AI-driven diagnostic methods?

Objective: to examine the practical, ethical and technical challenges faced in the field, with particular attention to issues of privacy, security and acceptance by patients and professionals.

5.1 Data source and search strategy

Scopus, a comprehensive citation database developed by Elsevier, was utilised for the preliminary analysis. Scopus provides access to peer-reviewed scientific outputs, with over 70 million bibliographic citations, abstracts, and bibliometric data across a wide range of academic disciplines, including natural sciences, engineering, medicine, social sciences, arts and humanities. It offers tools for analysing research trends, citations and academic collaborations, including metrics such as the h-index for quantifying the impact of authors and their publications.

Furthermore, we exported Scopus citations to VOS viewer, a bibliography management software, for further analysis.

To narrow the search to our fields of interest, we applied the following search query in the Scopus database:

("artificial intelligence" OR "machine learning" OR "deep learning") AND ("diagnosis") AND ("psychiatric disorders" OR "psychopathology" OR "neurological disorders"). The search yielded 1,245 articles. To refine the results, the following inclusion and exclusion criteria were applied:

5.2 Inclusion criteria

- Papers published between 2014 and 2024.
- Written in English.
- Focused on the application of AI diagnosis
- Peer-reviewed journal articles.
- For duplicate articles, the most recent version was included.

5.3 Exclusion criteria

- Non-journal articles.
- Duplicates where an older version existed.

After applying these criteria, the dataset was reduced to 651, covering various research fields. Figure 2 shows the distribution of the total material across subject areas. Subsequently, only articles related to the medical field were selected, resulting in a final database of 265 documents (Table 1).

From this refined dataset, the 10 most cited papers were identified for an in-depth analysis aimed at addressing the last 3 research questions.

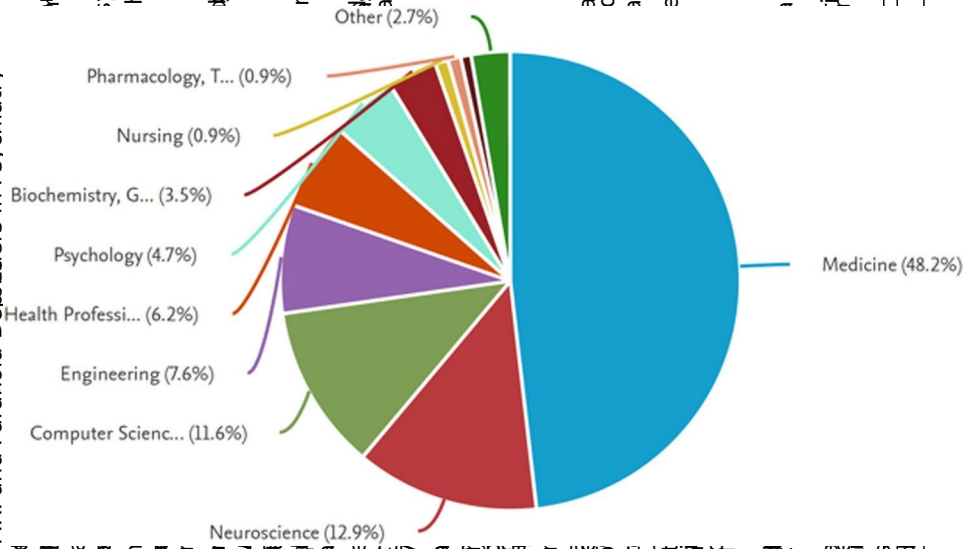
Authors	Year
	2024
	2024
	2024
	2024

Tab11 Complete table of the 265 papers belonging to the original database:

[illegible]

[illegible]

Fig. 2 Composition of Scopus articles based on the field area. The image is taken directly from the "Analyze Results" function of Scopus



5.4 Data presentation

The bibliographic information is presented in a structured manner, allowing for easy access to the full text of the articles. The default parameters for the search are set to include all articles published in the field of Medicine.

5.5 Data analysis

The analysis is based on the following criteria:

- Publication trends over time
- Geographical distribution of authors
- Main publication venues
- Classification of articles
- Algorithmic analysis

The top 10 most cited articles are listed below:

5.6 Evaluation metrics

To address the following challenges:

- Accuracy.
- Precision and recall.
- F1-score.
- Area under the curve.
- Sensitivity and specificity.

These metrics offer insight about how AI tools are validated in real-world diagnostic scenarios.

6 Results

In this section, the research results will be presented, providing an answer to the first four research questions.

RQ1 How many scientific studies have been published between 2014 and 2024 regarding the use of the use of AI-driven diagnostic methods?

This research question aimed to quantify the scientific community's interest in utilising AI for the early diagnosis of neurodegenerative diseases and psychopathologies. As depicted by Fig. 3, the number of publications within this domain has shown a consistent upward trajectory from 2014 to 2024. The number of publications rose from 4 in 2014 to 64 in 2024. This growth indicates a growing interest in the use of ML and DL to support early diagnosis of diseases that represent a significant burden on society.

RQ2 What are the most irrelevant research centers for studies on AI for the diagnosis of mental and neurological disorders?

This research question sought to identify leading research centers contributing to the dissemination of knowledge on AI applications in diagnosis. The findings, visualized in Fig. 4, highlight the institutions with the highest impact and publication output in this area. In particular, King's College is the leading institution with the highest number of publications.

RQ3 Which countries had the most active research centers?

This question focuses on the geographical distribution of research activity. As shown in Fig. 5 the United States leads in term of publication output, followed by China, the United Kingdom, Germany and Canada. Additional contributors include Italy, India, Australia, Turkey and Netherlands, underscoring a global effort in advancing AI for diagnostic purposes. Figure 6 represents a VOSviewer map of the collaboration between different countries in conducting studies.

RQ4 In which disorder is AI most commonly applied?

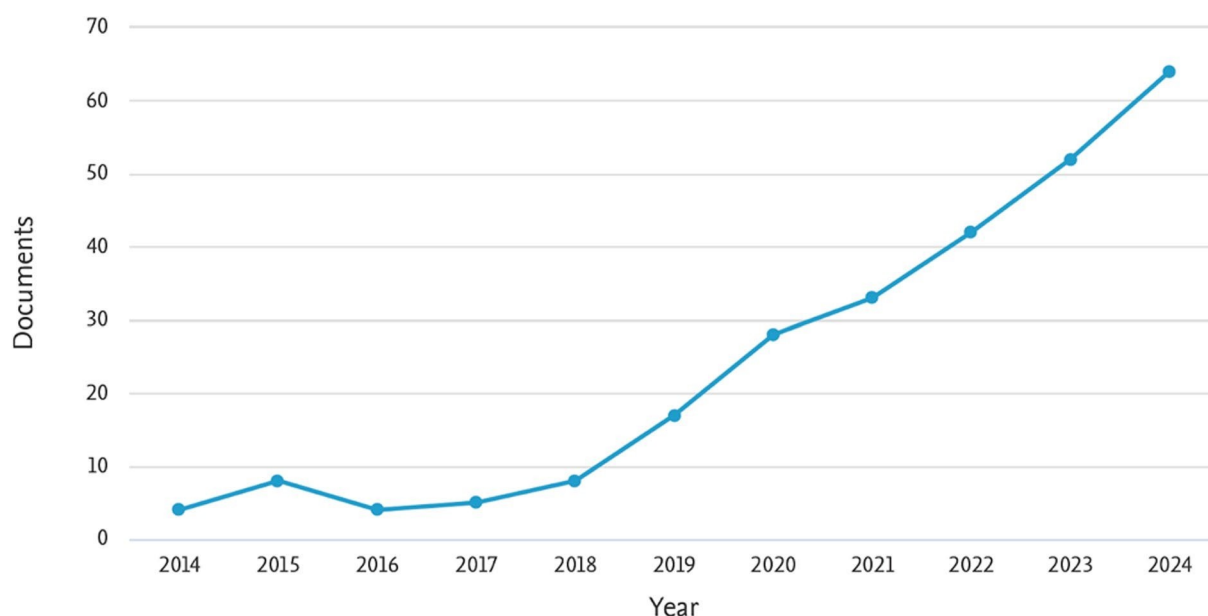


Fig. 3 Academics studies published from 2014 to 2024 in Scopus database. The image is taken directly from the “Analyze Results” function of Scopus

Fig. 4 Institutions where most of the research was conducted. At the bottom the number of articles is displayed, while on the left there is the name of the institute. The image is taken directly from the “Analyze Results” function of Scopus

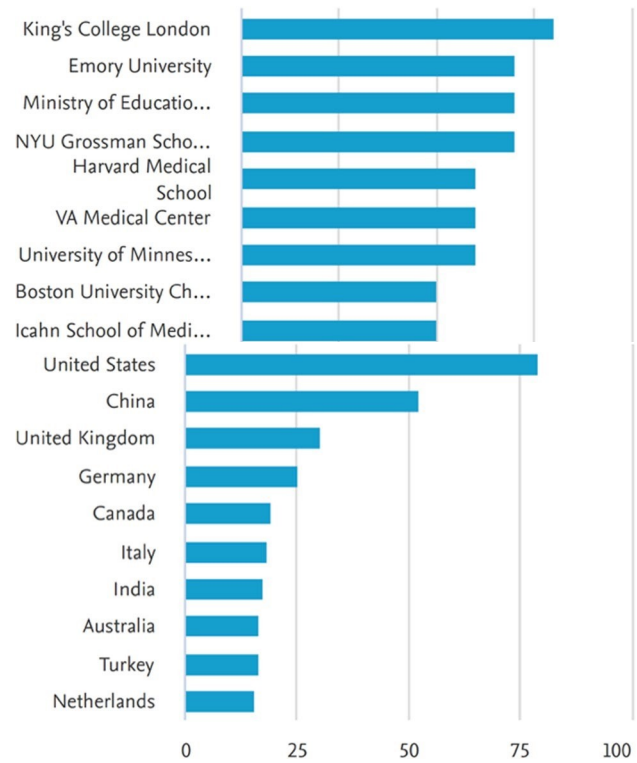


Fig. 5 Number of publications for each country. At the bottom there is the number of publications, while on the left there are the most involved countries. The image is taken directly from the “Analyze Results” function of Scopus

This research question aims to find the primary neurological and psychiatric diseases for which AI-driven diagnostic tools are frequently utilised. To conduct this investigation, we analysed the keywords from the 265 considered articles, as shown in Fig. 7.

Among them, there is a clear predominance of depressive disorder, followed by an equal distribution between various neurodegenerative diseases and psychiatric disorders. Bibliometric analysis of the co-occurrence of the keywords found in the index (Fig. 8) and co-authorship (Fig. 9) were then carried out using VOSviewer.

7 Analysis of the main papers

In this section, we analyse the 10 most cited articles from the final database, following to the selection criteria outlined in Sect. 5. A comprehensive overview is shared, highlighting the key areas of interest in the selected articles on the application of AI to diagnose both neurological and mental disorders. First, the most commonly used algorithms in this field will be listed and briefly explained, addressing research question number 5 (RQ5). Next, the metrics used to evaluate the performances of the aforementioned algorithms will be indicated (RQ6). Finally, the primary challenges faced in the selected articles will be discussed (RQ7).

In Fig. 10, the following information is given for each article: the title (first column), the year of publication (second column), the number of citations (third column), the method underlying the proposed technique (fourth column), and the dataset utilised for the analyses (fifth column). Where specific details are not provided, these are marked as “N/A”.

RQ5 Which AI algorithms have been most commonly used to diagnose both neurological and mental disorders?

This research question aims to offer professionals with a clear understanding of the algorithmic techniques applied in this field, as discussed in the selected studies. To this end, the technical aspects of each document are examined, including the algorithms employed and the datasets used. The most frequently employed techniques are outlined below:

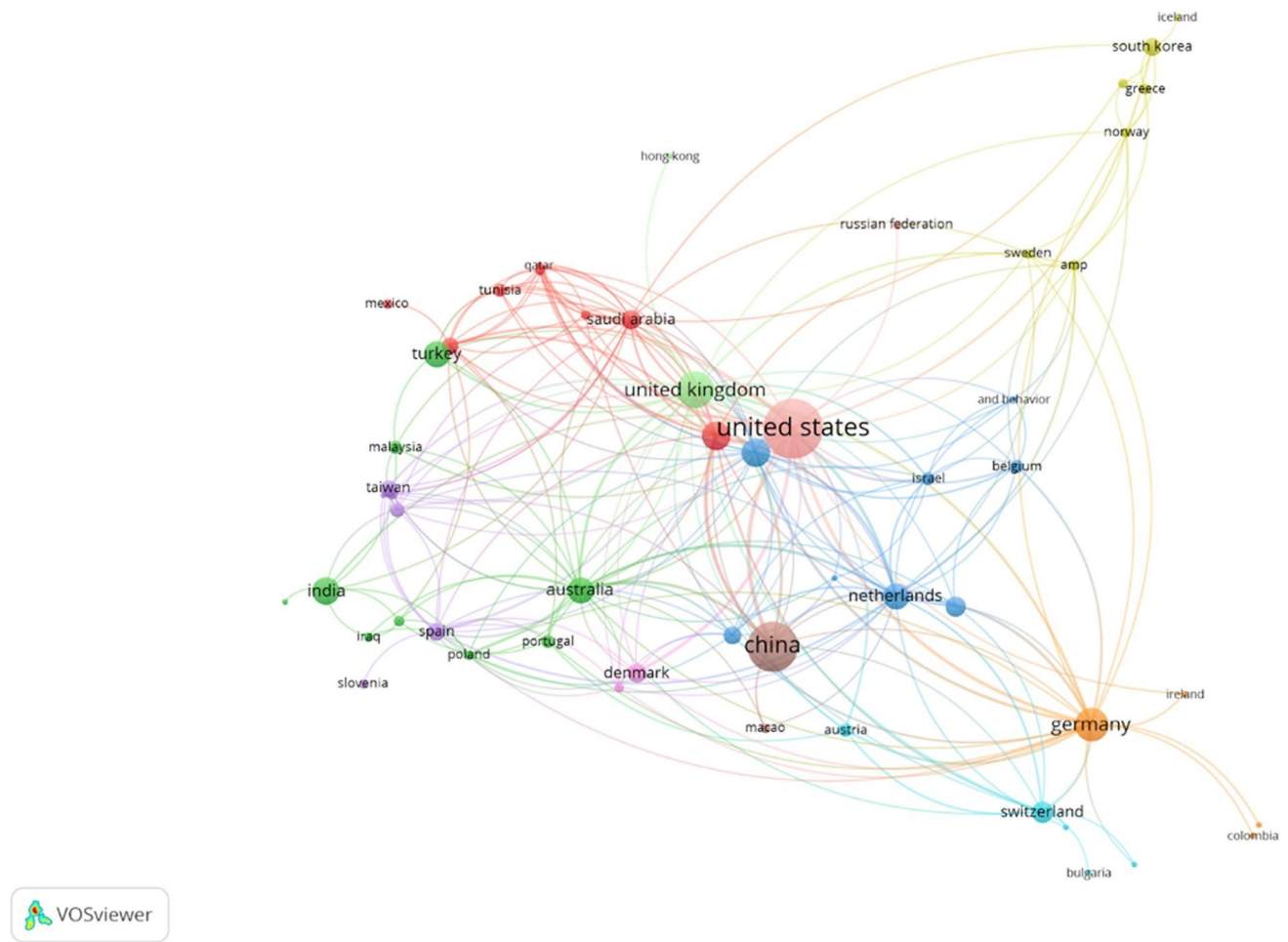


Fig. 6 Bibliometric analysis of collaboration between different countries in the publication of articles. The image was generated with VOSviewer

Fig. 7 Fields of application of AI in the diagnosis of psychiatric and neurological diseases, according to the Scopus database

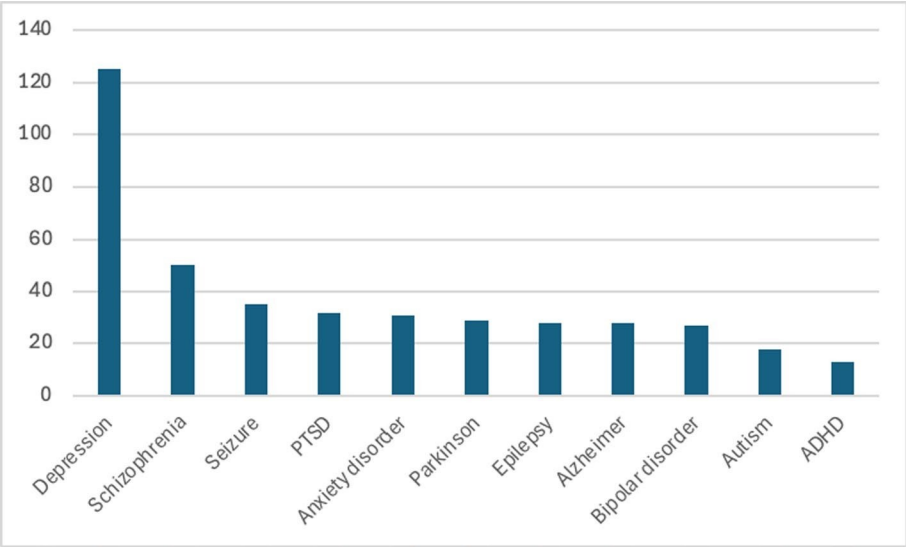


Fig. 8 Bibliometric analysis on the co-occurrence of the index keywords. Co-occurrence consists in the in the contemporary appearance of two keywords in the same article. In the diagram, each cluster is represented by a colour, and each element within a cluster shares the same colour. The size of the circle and label is determined by the frequency of the keyword's appearance. Each cluster groups keywords that describe an application domain of artificial intelligence in the medical field, and the lines connecting the elements represent the cooccurrence of keywords within an article

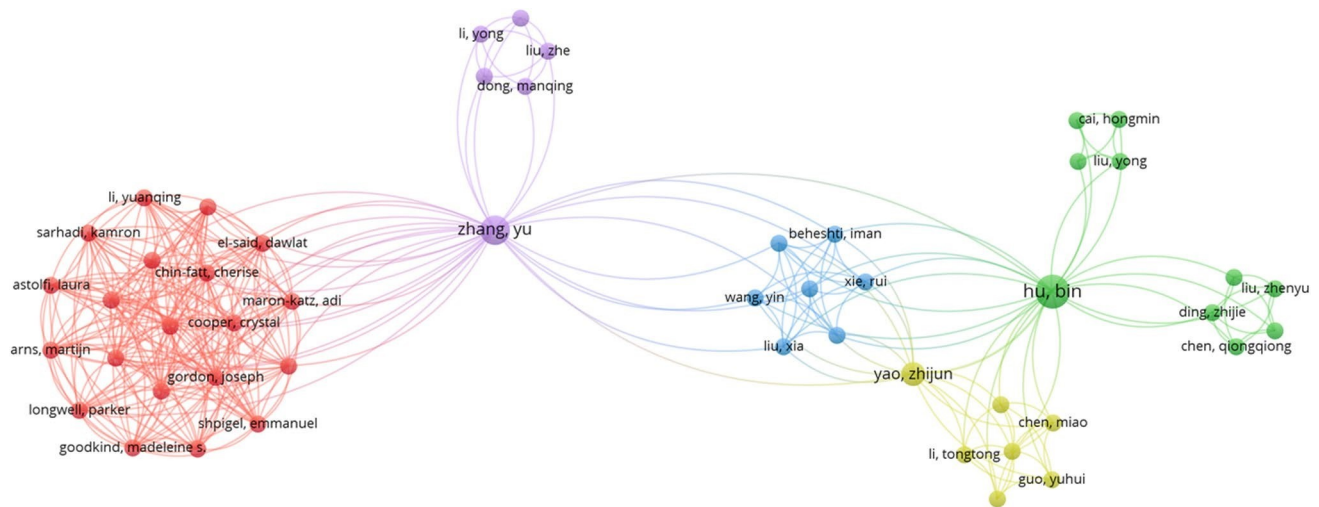


Fig. 9 Bibliometric analysis on the co-authorships generated with VOSviewer. The graph shows the collaboration relationships among authors of scientific articles with fewer than 25 co-authors. All articles were included, even those with only one citation, while authors without connections are not displayed. The nodes represent individual authors, and the size of the nodes reflects the number of associated publications. The lines indicate co-authorship between two authors. The different colors represent clusters automatically identified by VOSviewer, suggesting distinct research groups or communities with closer collaboration

Paper	Year	Citations	Methods	Dataset
Automated EEG-based screening of depression using convolutional neural network [45]	2018	447	CNN	Psychiatry Department, Medical College, Calicut, Kerala, India
Predicting suicides after psychiatric hospitalization in US army soldiers; The Army Study to Assess Risk and Resilience in Servicemembers (Army STARRS) [46]	2015	335	Regression trees, Elastic net regression	HADS (historical administrative data system) of the Army STARRS (army study to assess risk and resilience in servicemembers)
Unsupervised classification of major depression using functional connectivity MRI [47]	2014	153	MMC	First Affiliated Hospital of China Medical University and healthy
Use of machine learning to improve autism screening and diagnostic instruments: effectiveness, efficiency, and multi-instrument fusion [48]	2016	137	SVM	Balanced Independent Dataset
Bridging a translational gap: Using machine learning to improve the prediction of PTSD [49]	2015	121	SVM	Jerusalem Trauma Outreach and Prevention Study
Prevalence and Diagnosis of Neurological Disorders Using Different Deep Learning Techniques: A Meta-Analysis [50]	2020	111	CNN, DNN, RNN	N/A
Utilization of machine learning for prediction of post-traumatic stress: A re-examination of cortisol in the prediction and pathways to non-remitting PTSD [61]	2017	111	LGMM, SVM	N/A
Complex networks reveal early MRI markers of Parkinsons disease [54]	2018	105	Random forest, SVM	Parkinson's progression markers initiative (PPMI)
Identification of psychiatric disorder subtypes from functional connectivity patterns in resting-state electroencephalography [20]	2021	104	Sparse K-means	N/A
Linguistic markers predict onset of Alzheimer disease [55]	2020	101	SVM, logistic regression, naïve bayes classifiers	Framing heart study (FHS)

Fig. 10 Schematic summary of the content of the 10 most cited papers. The first column contains the title of the articles, the second the year of publication, the third the number of citations, the fourth the model used and the fifth the origin of the datasets used

Support vector machine (SVM):

SVM is a machine learning algorithm widely used for classification problems due to its simplicity and flexibility. It learns from examples to categorize or classify objects. Basically, the primary goal of SVMs is to determine a Hyperplane that effectively separates data points belonging to different classes. The chosen hyperplane is the one that maximises the distance between the hyperplane and the closest data points from each class. Recently, SVM has been employed to accurately predict diagnosis and prognosis of brain and psychiatric disorders, including Alzheimer's disease, schizophrenia and depression [36, 37]. **Convolutional neural network (CNN):**

A CNN is a type of artificial neural networks designed to simulate the human visual process. It consists of artificial neurons that handle inputs through various operations across multiple layers. In particular, it applies convolutional filters to input data to detect features and hierarchically learns more complex patterns through multiple layers. CNNs are widely used in machine learning due to their numerous applications, such as facial recognition, image and video analysis, handwriting recognition, anomaly detection, drug discovery and voice recognition [38].

Logistic regression:

Logistic regression is a statistical method used to predict binary outcomes (e.g., diseased/ healthy) based on one or more independent variables. It is widely used in healthcare research for decision-making models and disease state prediction. Logistic regression relies on maximum likelihood estimation and thus requires large sample size to ensure sufficient representation across outcome categories [39].

RQ6 What metrics were used to evaluate performance of the identified AI tools?

Paper	Metrics	Challenges
Automated EEG-based screening of depression using convolutional neural network [45]	Accuracy, sensitivity, specificity	Develop a self-learning model to diagnose depression through EEG analysis
Predicting suicides after psychiatric hospitalization in US army soldiers; The Army Study to Assess Risk and Resilience in Servicemembers (Army STARRS) [46]	AUC, concentration of risk (CR)	Develop a more effective and reliable algorithm to predict suicide risk among US army soldiers
Unsupervised classification of major depression using functional connectivity MRI [47]	Clustering and classification consistency	Develop an unsupervised ML approach using fMRI scans to improve objectivity of diagnosing MDD
Use of machine learning to improve autism screening and diagnostic instruments: effectiveness, efficiency, and multi-instrument fusion [48]	Unweighted average recall (UAR)	Create robust ML algorithms to enhance the accuracy and efficiency of ASD diagnoses
Bridging a translational gap: Using machine learning to improve the prediction of PTSD [49]	Accuracy, sensitivity, specificity, AUC	Enhance the personalization of PTSD risk prediction using ML, to analyze less obvious predictive factors
Prevalence and Diagnosis of Neurological Disorders Using Different Deep Learning Techniques: A Meta-Analysis [50]	Accuracy, sensitivity, specificity	Review the use of DL techniques for early diagnosis of neurological and psychiatric disorders
Utilization of machine learning for prediction of post-traumatic stress: A re-examination of cortisol in the prediction and pathways to non-remitting PTSD [51]	Accuracy, sensitivity, specificity, positive predictive value, AUC	Develop predictive models to classify individuals at risk for PTSD through neuroendocrine responses
Complex networks reveal early MRI markers of Parkinson's disease [54]	Accuracy, sensitivity, specificity, AUC	Propose a novel early diagnosis strategy for detecting Parkinson's disease prior to motor symptoms onset
Identification of psychiatric disorder subtypes from functional connectivity patterns in resting-state electroencephalography [20]	Accuracy, sensitivity, AUC	Identify subtypes of psychiatric disorders using ML techniques to analyze patterns from EEG data
Linguistic markers predict onset of Alzheimer disease [55]	Accuracy, sensitivity, specificity, positive predictive value, AUC	Predict the future onset of Alzheimer's disease in healthy individuals using automated linguistic analysis

Fig. 11 Schematic representation of the results of the 10 most cited papers' analysis. The first column presents the papers' titles, the second the metrics employed to evaluate the models' performances and the third summarises the challenges addressed by the studies

The aim of this research question is to offer an overview of the metrics used to evaluate the performance of the algorithms extracted from the 10 selected articles. The second column of Fig. 11 lists the metrics associated with the respective studies.

The most commonly employed metrics include the following:

Accuracy:

Accuracy measures the predictive ability of a classification algorithm on the testing data, independent of decision biases or prior probabilities, thus allowing the performance of multiple systems to be compared on a common, interpretable scale. Valid and precise assessments of intrinsic accuracy enable users to determine the reliability of diagnostic tools. [40, 41] **Area Under the Curve (AUC):**

AUC evaluates a classification algorithm's ability to produce probability estimates rather than solely class predictions. Thus, this metric is more sensitive than accuracy because it is independent of decision thresholds and class distributions. AUC has therefore been shown to be a superior discriminative indicator in numerous applications and statistical tests [42].

Sensitivity:

Sensitivity, also known as the true positive rate, represents the probability of obtaining a positive result among individuals who actually have the target condition. It reflects the ability of a test to detect true positives (e.g., diseased individuals) [43, 44].

$$TP/(TP + TN)$$

Specificity:

Specificity, also known as the true negative rate, represents the probability of obtaining a negative result among individuals who do not have the target condition. It indicates the ability of a test to detect true negatives (e.g., healthy subjects) [43, 44].

$$TN/(TN + FP)$$

Positive predictive value (PPV):

PPV represents the likelihood that a positive test result corresponds to the target condition. Furthermore, by indicating the portion of true positives among positive results, it helps assess a test's comparability to the 'gold standard' [43, 44].

$$TP/(TP + FP)$$

The metrics described above are those utilised in the 10 articles extracted that underwent in-depth analysis. Consequently, it is possible that other metrics may be used in additional studies within the database, depending on the context and objectives of the research.

RQ7 What challenges have been addressed in the development and implementation of AI-driven diagnostic methods?

The integration of AI in the medical field has challenged the traditional approaches to patient care, encompassing diagnosis, treatment and management of pathologies. However, the advent of personalized care has encountered obstacles, including ethical concerns, technical challenges, privacy-related issues, and acceptance by patients and healthcare professionals. To emphasize the importance of the challenges associated with implementing AI-driven diagnostic tools, we analysed the 10 most cited articles from the identified database. These articles were examined to determine how they address specific challenges in the context of AI applications to neurological and mental disorders. In the third column of Fig. 11, the challenges discussed in these articles are summarised in relation to the specific goals and context of each document. Detailed descriptions of the articles are given below:

Acharya et Al. [45] tackles the challenge of diagnosing depression, a condition that often goes undetected despite the availability of effective treatments. The article presents a self-learning model that can detect distinctive features in EEG data. The model was trained and tested on a self-made dataset made by EEG data collected from 15 healthy individuals and 15 depressed patients. The sampling rate of the signals was 256 Hz with a notch filter of 50 Hz to eliminate power line interference. The final dataset had 4348 records divided in half across the two populations of subjects. By creating a new dataset ad hoc for this task, the authors guaranteed a high level of data quality and heterogeneity. Subtle differences in brain activity between depressed and non-depressed individuals are identified through EEG analysis. In particular, EEG

signals from the right hemisphere are more distinctive for diagnosing depression. This approach offers a clear advantage in terms of efficiency and improves sensitivity, specificity and accuracy of the diagnosis.

Kessler et Al. [46] explores a new approach to assess suicide risk among US Army soldiers within 12 months following treatment for a psychiatric disorder. The study addresses the challenge of accurately predicting suicide risk, a task that remains difficult despite known risk factors. The authors aim to overcome this limitation by developing a more effective and reliable risk prediction algorithm. To train and test it, the authors built their dataset starting from the Historical Administrative Data System of the Army Study to Assess Risk and Resilience in Servicemembers. This dataset had missing values, and it was inconsistent in some components probably because it wasn't created for research purposes. To ensure the reliability of the outputs, the remaining missing values were resolved using randomly selected multiple imputations and the inconsistencies were fixed with rational imputations.

Zeng et Al. [47] focuses on overcoming the limitations of traditional psychiatric diagnoses, particularly for Major Depressive Disorder (MDD), which often rely on self-reported symptoms and clinical observations that are susceptible to bias. The authors develop an unsupervised machine learning approach using fMRI scans, enabling classification without pre-labeled data, thus enhancing diagnostic objectivity. To do so, an ad hoc dataset was created, collecting the imaging scans of 24 patients diagnosed with Major Depressive Disorder and 29 healthy subjects. The healthy controls were selected on the basis of demographic similarity to each depressed patient. To avoid biases in the output, all patients underwent MRI under the same conditions and the scans were acquired and processed in the same manner. This research supports the application of machine learning in clinical practice, specifically reducing diagnostic bias and improving clinical outcomes.

In Bone et Al. [48] the objective is to create new diagnostic tools for Autism Spectrum Disorder (ASD) using ML techniques. The research aims to improve the performance of caregiver-report instruments to achieve the “goldstandard” diagnosis. The study seeks to develop robust ML algorithms that achieve efficiently even with datasets containing conflicting data, thereby contributing to more accurate and efficient diagnostic assessments for ASD. The dataset involved in the experiment included ADI-R and SRS scores for 1,264 verbal individuals with ASD and 462 with non-ASD developmental or psychiatric disorders and the subjects' data were drawn from an IRB approved repository.

Kartsoft et Al. [49] addresses the challenge of personalising PTSD diagnosis using ML techniques to analyse various combinations of predictive features. Unlike previous studies that focused on predictive factors in large groups, this research prioritises less obvious and less frequently recorded variables. The authors apply ML to a large dataset, utilizing support vector machines (SVM) to predict persistent PTSD symptoms. The dataset was made of data collected for the Jerusalem Trauma Outreach and Prevention Study. To predict PTSD symptoms trajectories, features about event characteristics, emergency department records and early symptoms were collected for 957 trauma survivors. This study aims to enhance the accuracy and personalization of PTSD risk prediction.

Gautam et Al. [50] aims to fill a gap in the existing literature by examining the use of DL techniques for the early diagnosis of neurological and psychiatric disorders, including cerebrovascular disease, Alzheimer's, Parkinson's, epilepsy, cerebral palsy, multiple sclerosis, autism and migraine. Given the complexity and severity of these diseases, which are often chronic and have poor prognosis, timely and accurate diagnosis is crucial. The article introduces various DL techniques and presents the major neurological disorders, analyzing the publication trends related to these conditions.

Galatzer-Levy et Al. [51] explores the use of advanced computational approaches to develop predictive models capable of classifying individuals with heterogeneous risk factors. In particular, they used the data collected by Shalev et al. (2008) [52] and Videlock et al. (2008) [53], two parts of a longitudinal study that, using classical GLM statistics, previously failed to show a group-wide association between endocrine markers at hospitalisation time and PTSD status 5 months later. The data included assessment of trauma exposure, personal information, neuroendocrine and psychiatric assessments of 152 patients. Notably, the neuroendocrine response, specifically cortisol, emerges as a stable predictor of the development of post-traumatic stress disorder (PTSD) when combined with other clinical data. The article emphasises early prediction and risk factor identification, suggesting that manipulating the mechanisms underlying the development of disorders can help prevent their onset.

Amoroso et Al. [54] propose a novel strategy for early diagnosis of Parkinson's disease (PD) prior to the manifestation of motor symptoms combining neural network and clinical features. The diagnostic tool is based exclusively on markers derived from MRI and uses an unsupervised methodology to model brain activity in both healthy subjects and patients to explore the brain areas most affected by the disease. To do so, the authors relied on data from the Parkinson's progressive markers initiative, both for clinical and imaging data. This online repository is strongly research oriented and

very reliable. However, they do not specify the solutions adopted in the case of empty cells or inconsistent measurements. The author's approach focuses on the identification of a combination of different markers, enabling accurate early diagnosis and monitoring of disease progression.

Zhang et Al. [20] aims to identify existing subtypes of psychiatric disorders, such as post-traumatic stress disorder (PTSD) and major depressive disorder (MDD). The identification of these subtypes is conducted through an approach based on functional connectivity patterns, detected via resting-state EEG, and the application of machine learning techniques to identify solutions for connectivity-based diagnosis. To ensure output reliability, the data used to train and test the algorithms were collected specifically for this experiment.

In Eyigoz et Al. [55] aims to predict the future onset of Alzheimer's disease in cognitively normal subjects through automated linguistic analysis. The study used linguistic data to create predictive models and analyse the correlation between lower early-life linguistic performance and higher incidence of cognitive decline. The experiment relied on data collected by the Framingham heart study, a longitudinal study that tests cognitive status and its decline since 1975. Also in this case, the data source is reliable, but the authors did not reveal how they targeted inconsistent measurements. It emerges that language performance can reveal early signs of cognitive decline and thus be used as markers to identify at-risk individuals, enabling timely interventions.

8 Discussion

This document has investigated the scientific community's interest in the implementation of AI techniques, particularly machine learning, in the diagnosis neurological and mental disease over the past decade.

Our review of publications from 2014 to 2024 reveals a significant increase in research activity in this area. Specifically, the number of publications has risen from a minimal count to a substantial figure by 2024, highlighting the growing interest in using ML to support the early diagnosis of diseases.

The analysis focused on a total of 265 documents, selected based on inclusion and exclusion criteria, focusing exclusively to the medical domain. This final dataset was used to examine the evolution of publications, dissemination channels, the countries that contributed the most, and patterns of collaboration across research networks.

Lastly, to conduct a more in-depth analysis of our research, we selected the 10 most-cited articles, addressing the research questions defined previously.

The analysis underscores the growing interest of professionals in the development of new diagnostic methodologies. It is evident that AI is an important ally of the medical field in reducing the burden that these diseases impose on society. In particular, the results show that these technologies are help anticipate the diagnosis of the discussed pathologies.

Regarding neurodegenerative diseases, such as Alzheimer's, Parkinson's, and others, the primary objective is early diagnosis before the onset of debilitating symptoms. Early identification can prolong the patient's life enhancing its quality, reduce the burden on caregivers, and lessen the strain on the healthcare system. Although these diseases are chronic and progressive, it has been proven that the use of algorithms, which are fed clinical data, can predict the likelihood of disease onset and promote early management strategies.

In the field of psychiatric disorders, AI aims to improve quality of life and slow disease progression. Moreover, these algorithms have also proven crucial in identifying risk factors that may lead to severe consequences. In the case of posttraumatic stress disorder, these techniques have been able to identify warning signs for suicide risk, allowing for the monitoring of individuals and preventing tragic outcomes.

While the use of AI in diagnosis offers great potential, it also presents complex challenges and issues. The first hurdle to overcome is the acceptance of these technologies within the diagnostic process, both by healthcare professionals and, most importantly, by patients. Resistance to these technologies may be caused by a lack of trust or familiarity with their capabilities. Secondly, there are ethical and legal concerns, especially regarding accountability in the event of errors and the protection of patient privacy, given the vast volume of sensitive data required to enable these technologies to do their tasks.

Finally, data reliability and quality are also crucial challenges. The models are often trained on incomplete, biased, or low-quality data, which can lead to inaccurate or misleading diagnoses. Ensuring the integrity of training data is essential to the safe and effective use of these technologies.

At the end of this analysis, it is possible to state that the benefits of using these techniques are evident, while there are equally significant controversial aspects to consider. Rigorous approaches and close collaboration between developers, healthcare professionals and policymakers are essential to ensure AI as a supportive and safe tool in medical diagnosis.

9 Limitations and future directions

Despite the clear potential of artificial intelligence in the early diagnosis and management of psychiatric and neurological disorders, several critical limitations remain. First, is the issue of poor generalizability of the models, which are often developed using narrow and homogeneous datasets. This significantly reduces their applicability in real-world, heterogeneous clinical settings. Another, major challenge is interpretability, as many algorithms operate as “black boxes”, making it difficult for healthcare professionals to understand, and ultimately trust the decisions suggested by these systems. Additionally, the presence of bias in the training data can compromise both the fairness and reliability of the models.

To address these issues, future developments must focus on building large, shared, and representative dataset, as well as on developing interpretable models, for example through the use of explainable AI techniques. Although, none of the 10 most cited paper has implemented these kinds of techniques, recent studies have begun to embrace this direction, demonstrating how XAI can bridge the gap between model complexity and clinical interpretability, creating better healthcare solutions and enhancing models' accuracy [56]. For instance, Chadaga et al. (2023) [57] used advanced machine learning methods combined with explainable AI techniques to predict the efficacy of hematopoietic stem cell transplants in pediatric patients. The final model achieved excellent accuracy, and the use of XAI methods made it possible to explain predictions by identifying the most relevant variables, making the system interpretable and useful for supporting clinical decisions.

Similarly, another study applied multiple explainable AI approaches to predict stroke risk. XAI techniques highlighted key variables such as age, BMI, hypertension, average blood glucose levels, and heart disease [58].

Again, Wani et al. (2024) [59] proposes a novel framework to classify subjects with cardiovascular diseases. This approach enhances DL's accuracy by integrating a XAI model that provides comprehensive explanations of the output at a global and local level. As a result, the model shows increased performances as compared to already existing models for this task, while assisting medical practitioners with their diagnostic procedure. Following the same approach, XAI has been proven useful in other branches of medicine, such as oncology. For instance, combining the efficacy of NNs with XAI explanations was demonstrated convenient for lung cancer detection [60] and to enhance breast cancer classification [61].

In the context of Alzheimer's detection, Goenka et al. (2024) [62] presented a deep learning-based classification model using 3D MRI images, which makes the process fully automated without the need for manual feature engineering. The system demonstrated notable accuracy in binary classification between affected patients and healthy controls, proving to be highly promising for clinical support in early Alzheimer's diagnosis.

Lastly, Goswani et al. (2024) [63] combined a deep neural network with XAI (Grad-CAM visualizations) to detect sickle cell disease from microscopic blood images, enhancing diagnostic precision while ensuring interpretability through visual cues. These examples illustrate how explainable AI can transform complex models into transparent, decision support tools in healthcare. Furthermore, promoting clinical studies will be essential to facilitate the integration of these new technologies into healthcare environments, with the goal of supporting, not replacing, clinical judgment.

The limitations highlighted by this research undoubtedly represent threats to validation. In particular, external validity is undermined, which poses a major issue: models fail to generalize to different clinical contexts or patient populations, limiting their applicability beyond the original study setting. This directly leads to a lack of clinical validation, as models that are not tested in real-world settings may not perform as expected in daily clinical practice, and it becomes impossible to determine whether they truly predict what they are intended to. Moreover, heterogeneous, poorly standardized data and the presence of bias can distort relationships between variables, introduce uncontrolled confounding factors, and threaten internal validity.

10 Conclusion

Our analysis highlights a growing use of AI-driven diagnostic methods. The impact of these new approaches has been evident in the last decade in both psychiatric and neurological diseases. Several methods have been proven useful from different points of views. Early diagnoses and risk factor identification are just two of the multiple challenges that can be addressed with AI-driven techniques. These new revolutionary approaches could lead to improve the intervention timing

and to increase the quality of the patients' life, while reducing the caregiver's burden. A more personalized medicine and a more accurate and faster diagnosis are goals that emerge to be targeted by these recent developments. In this way, the healthcare system could improve the patient management and anticipate the treatment of a disease, avoiding tragic outcomes. However, alongside these promising developments, future research must investigate and address the limitations that have emerged from our research. Data reliability, privacy compliance and the presence of bias in the datasets are key aspects to target to enhance model reliability and to avoid inaccurate diagnoses. Furthermore, topics such as acceptability and explainability also need to be thoroughly investigated. Addressing these topics is essential to foster trust and confidence in AI-driven diagnostic methods. Only by overcoming these limitations will it be possible to fully realise the advantages that the development of AI in diagnostic can offer.

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Data Availability Our data have been freely downloaded from Scopus using the keywords mentioned in the paper. The dataset used in our analysis can be easily replicated: <https://www.scopus.com/search/form.uri?display=basic>.

Declarations

Ethics approval and consent to participate Not applicable.

Consent for publication Not applicable.

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